

IT9205 - Deep Learning for Computer Vision	
Project Topic Proposal Form	
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Date	4/3/26
Problem Domain	Automated Vehicle Detection and Classification in Traffic Scenes
	e.g., Medical Imaging, Traffic Analysis, Agriculture, Sports Analytics, etc.
Dataset Name	Vehicles-COCO Dataset (MSCOCO Vehicle Detection)
	Name of the dataset you plan to use.
Dataset Source (URL)	https://universe.roboflow.com/vehicle-mscoco/vehicles-coco
	Direct link to the dataset (Kaggle, Hugging Face, Roboflow, etc.)
Dataset Size & Details	Total Images: 18,998 annotated images Train/Test Split: Pre-split training and validation sets (approximately 80/20) Vehicle Classes: 4 main categories - Car (sedans, SUVs, hatchbacks) - Bus (public transportation vehicles) - Truck (commercial delivery trucks) - Motorcycle (two-wheelers, scooters) Image Resolution: Varying resolutions from 400x300 to 1920x1080 pixels Annotations: - Bounding boxes for each vehicle instance - Class labels for each detection - Multiple vehicles per image supported - Available formats: COCO JSON, Darknet TXT, Pascal XML, TFRecord Quality: Professional, real-world street and traffic scenes with diverse locations, lighting conditions (day/night), weather conditions (sunny/rainy), and viewing angles. Includes urban streets, highways, parking lots, and intersection footage. Pre-labeled: YES - All images include ground-truth bounding box annotations and class labels Approximate size, number of images, number of classes/categories, image resolution.
Alternative Dataset (Optional)	Name: Vehicle Detection Dataset (Roboflow) URL: https://universe.roboflow.com/lynkeus03/vehicle-detection-by9vs Size: 9,200+ images with 4 vehicle classes (car, bus, truck, motorcycle) Backup Purpose: Alternative dataset if primary has compatibility issues Name and URL of a backup dataset in case the primary is not suitable.
Task 1: Classification	What will you classify? Classify vehicle types in images into 4 categories: 1. Car (sedans, SUVs, hatchbacks, family vehicles) 2. Bus (public transportation vehicles, large passenger buses) 3. Truck (commercial delivery trucks, cargo vehicles) 4. Motorcycle (two-wheelers, scooters) The classifier receives an image and outputs: (1) vehicle type prediction, (2) confidence score Why is it meaningful for this domain? 1. Traffic Monitoring: Cities need to analyze vehicle composition for urban planning and congestion analysis. 2. Smart Parking: Parking systems need to identify vehicle types to direct drivers to appropriately-sized spots (motorcycles need less space than trucks). 3. Toll Collection: Highway toll systems need to classify vehicles to charge appropriate fees (motorcycles pay less than trucks). 4. Autonomous Vehicles: Self-driving cars must classify other vehicles to predict their behavior (trucks move differently than motorcycles). 5. Traffic Safety: Emergency response systems need to identify vehicle types for dispatch and collision prediction. 6. Urban Planning: City planners analyze vehicle composition on roads to design appropriate infrastructure (bike lanes vs truck lanes). 7. Real-world Value: Traffic classification is a core component of smart city systems processing millions of vehicles daily. What will you classify? Why is it meaningful for this domain?
Task 2: Detection or Segmentation	Task Chosen: Object Detection (not Segmentation) What will be detected? Detect and localize all vehicles in a scene image using bounding boxes: - Identify precise location of each vehicle (bounding box coordinates) - Classify the vehicle type for each detection - Handle multiple vehicles in single image - Estimate confidence score for each detection Example: Street scene with 5 vehicles → Output 5 bounding boxes with types (2 cars, 1 bus, 1 truck, 1 motorcycle) How does it relate to Task 1? HIERARCHICAL INTEGRATION (End-to-End Pipeline): Task 1 (Classification) → Classify vehicle type from image region Task 2 (Detection) → Locate WHERE each vehicle is in image Together they enable complete automation: Step 1 - Detection: Find all vehicles in image Input: Street scene image Output: Bounding boxes at vehicle locations Step 2 - Classification: Identify vehicle type at each location Input: Bounding box region Which task was chosen (detection/segmentation)? What will be detected or segmented? How does it relate to Task 1?
For Instructor Use Only	
Status	Approved
Instructor Feedback	Dataset and tasks are suitable, clear improvement since the previous proposal, well done. Keep your deliverables (report, notebook explanations) concise and focused on the technical task instead of moving away from the problem domain. A suggestion for Part F: consider model optimization for edge deployment, which is relevant to real-time traffic monitoring on roadside cameras, that are considered edge devices with limited computation. There should be a clear tradeoff analysis between accuracy and inference speed.